Success Rate Model

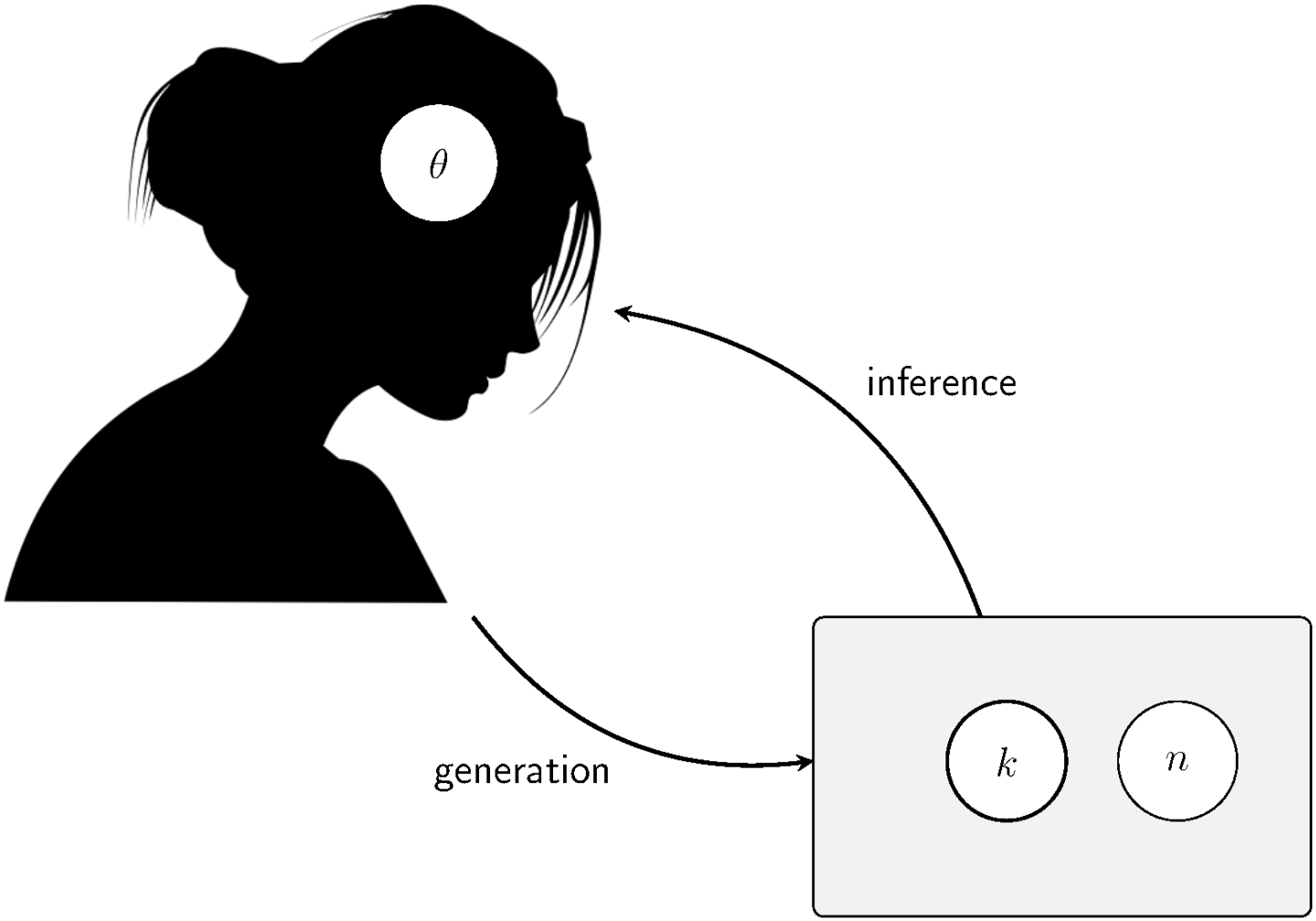
Michael D. Lee

Success Rate Model

- A common form of behavioral data is
  - a set or sequence of binary outcomes
  - one outcome corresponding to a success and the other to a failure
- Some examples
  - right or wrong response to each of a set of true-or-false questions
  - series or correct and incorrect responses in a visual identification task
  - participants in a memory task either succeeding or failing to recall items
- The success rate model assumes the probability of a success is the same throughout
  - no learning, no sequential effects

Success Rate Model Structure

- The success rate model has the structure
  - observed data k successes out of n observations
  - latent parameter for the success rate θ



Coin Tossing

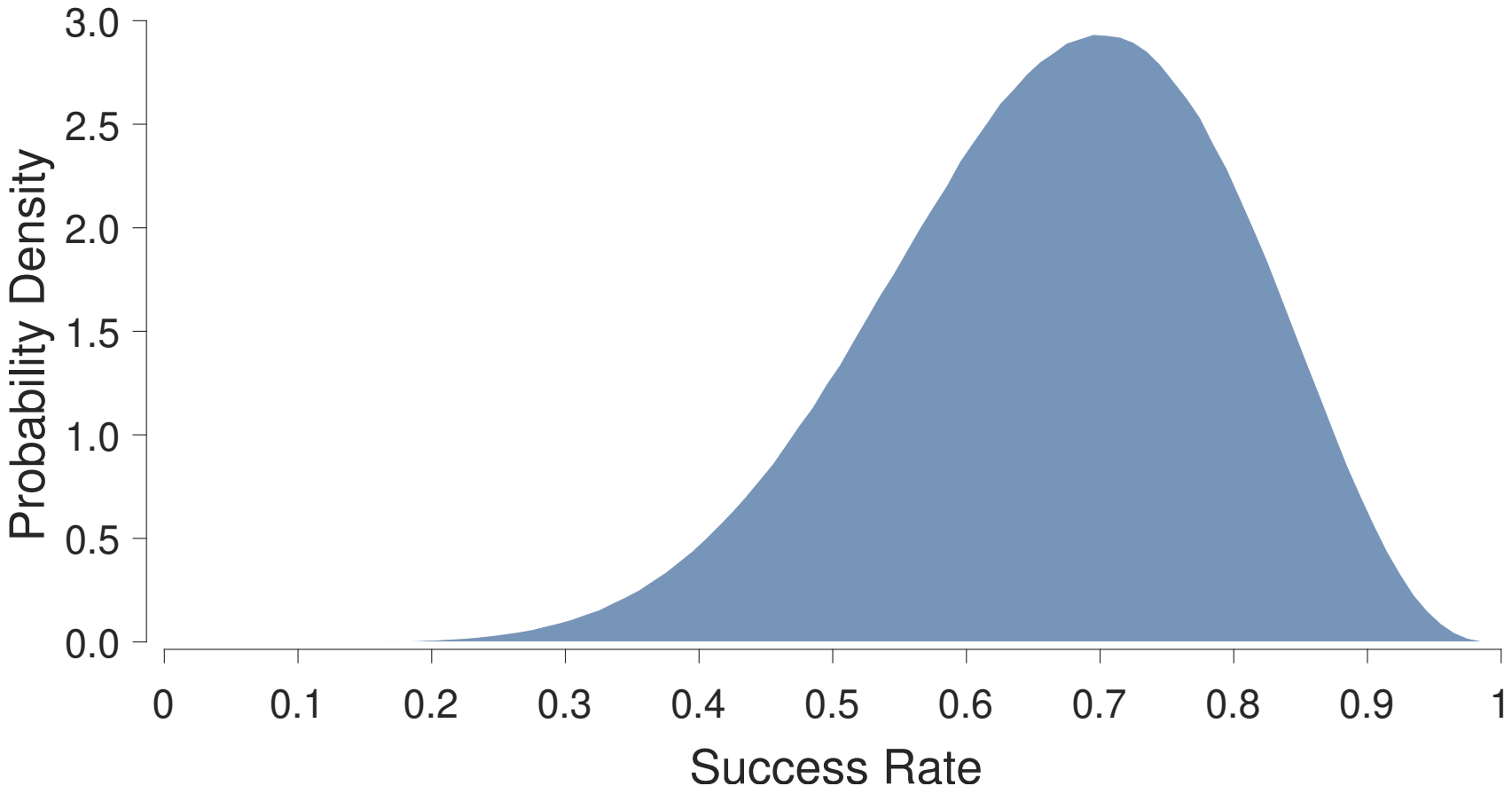
Coin Tossing

- Bartoš et al. (2023) tested the hypothesis that a coin toss is slightly more likely to land on the side of the coin that was facing upwards when it was tossed
  - they found evidence for this hypothesis based on 350,757 tosses

- We consider data just from the first 102 tosses, all of which started with tails facing upwards
  - the first 10 tosses yielded 7 tails
  - the first 102 tosses yielded 52 tails

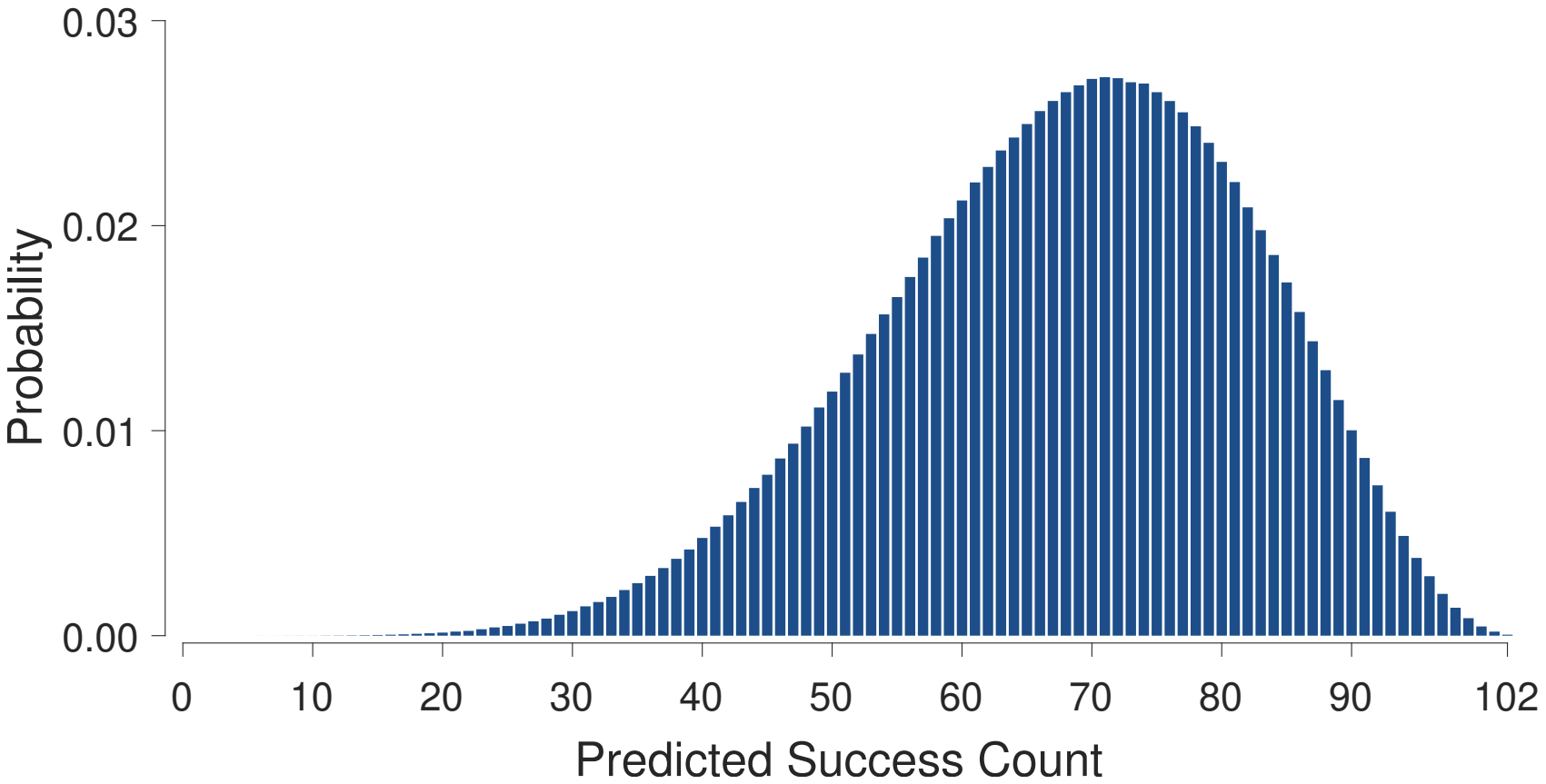
Success Rate Posterior

- The posterior distribution of θ for $k = 7$ successes from $n = 10$ tosses
  - rates around 0.7 are the most likely, but the rate could be as high as 0.95 or as low as 0.25
  - the 95% credible interval for θ is $(0.39, 0.89)$



Success Count Posterior Prediction

- The posterior predictive distribution of $\hat{k}$ successes for $\hat{n} = 102$ tosses
  - counts around 70 are most likely, but it is possible the outcome is fewer than 30 or greater than 100 successes
  - the 95% credible interval for $\hat{k}$ is $(38, 92)$



Student Grades

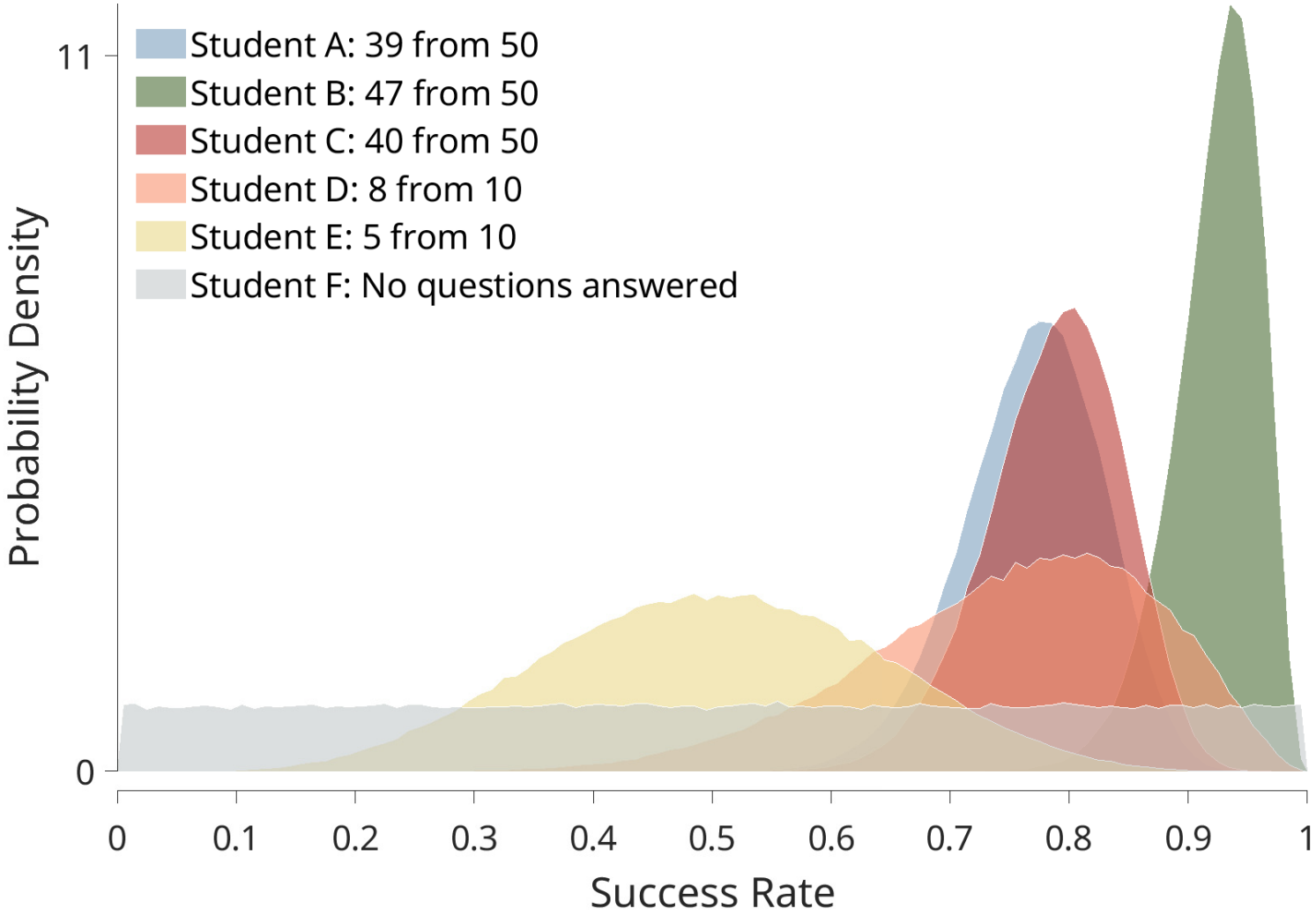
Student Grades

- A class of students are taking a fifty-item exam
  - students A–C have completed the exam
  - students E and F have answered the first 10 questions
  - student F has not started
- The success rate model can be applied by treating correct answers to questions as successes

Student	Correct	Completed	Percentage
A	39	50	78%
B	47	50	94%
C	40	50	80%
D	8	10	80%
E	5	10	50%
F	0	0	–

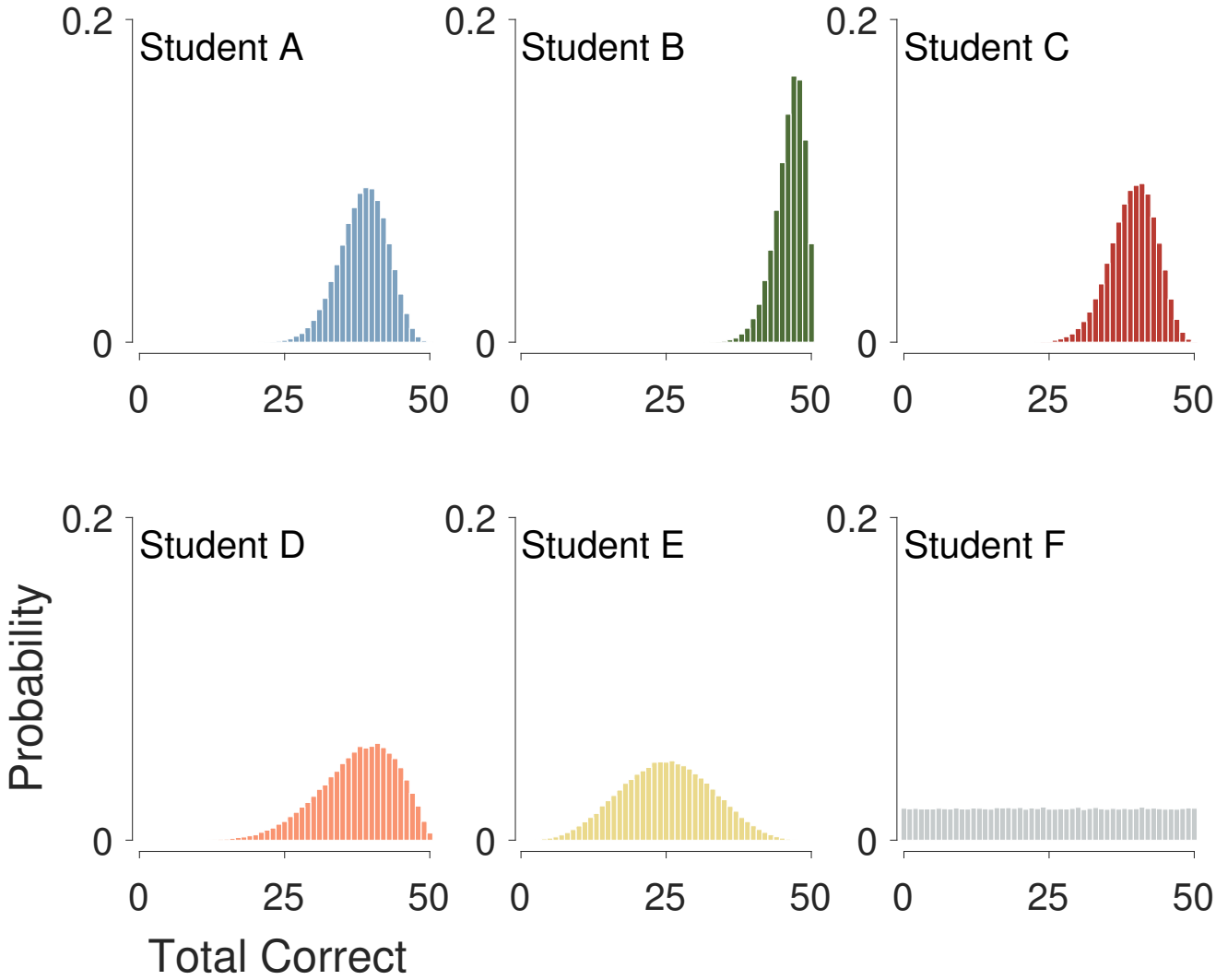
Success Rate Posteriors

- The certainty of the posteriors reflects the number of questions completed
  - for the extreme case of Student F, the posterior is the prior



Success Count Posterior Predictions

- The posterior predictions are for the number of correct questions $\hat{k}$ once the exam is completed with $\hat{n} = 50$



References i

Bartoš, F., Sarafoglou, A., Godmann, H. R., Sahrani, A., Leunk, D. K., Gui, P. Y., ... others (2023). Fair coins tend to land on the same side they started: Evidence from 350,757 flips. *arXiv preprint arXiv:2310.04153*.